JOAO P. A. DANTAS

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EDUCATION

Aeronautics Institute of Technology, São José dos Campos, Brazil — *Jan 2019 – Jun 2025*
Ph.D. in Electronic and Computer Engineering
Cum. GPA: 9.5 / 10.0

Carnegie Mellon University, Pittsburgh, PA — *Nov 2021 – Jan 2023*
Ph.D. Exchange Program in Robotics

Aeronautics Institute of Technology, São José dos Campos, Brazil — *Aug 2016 – Jan 2019*
M.Sc. in Electronic and Computer Engineering
Cum. GPA: 8.5 / 10.0

Stony Brook University, Stony Brook, NY — *Jul 2014 – Jul 2015*
One-year Undergraduate Exchange Program in Mechanical Engineering
Cum. GPA: 3.93 / 4.00

Aeronautics Institute of Technology, São José dos Campos, Brazil — *Jan 2011 – Dec 2015*
B.Sc. in Mechanical-Aeronautical Engineering
Cum. GPA: 8.7 / 10.0

RESEARCH EXPERIENCE

SETMV — Virtual Military Shooting Range Simulator — Jul 2026 – Present
Department of Aerospace Science and Technology (DCTA) — *São José dos Campos, Brazil*
Virtual reality and 3D simulation for military shooting instruction, with performance analytics comparing simulated training with live-fire benchmarks to reduce training costs and ammunition usage.

C2 — Command and Control through Aerospace Data Fusion — Mar 2025 – Present
Institute for Advanced Studies (IEAv) — *São José dos Campos, Brazil*
Advanced Command and Control (C2) frameworks leveraging massive aerospace data fusion and high-performance computing to support operational planning in complex defense scenarios.

ASA — Aerospace Simulation Environment — Dec 2020 – Present
Institute for Advanced Studies (IEAv) — *São José dos Campos, Brazil*
Evaluation of military operational scenarios and tactical protocols for the Brazilian Air Force using the custom-built, object-oriented **ASA** simulation framework.

Co-development of a Bespoke Air-to-Air Weapon Simulation Environment — Mar 2024 – Dec 2025
Institute for Advanced Studies (IEAv) — *São José dos Campos, Brazil*
Offset-agreement cooperation between IEAv and MBDA to co-develop an air-to-air weapon simulation environment integrated with IEAv simulation capabilities and to train the Brazilian team.

Safe and Seamless Operation of Manned and Unmanned Aircraft — Nov 2021 – Jan 2023
Robotics Institute, Carnegie Mellon University — *Pittsburgh, PA*
Worked with **Prof. Dr. Sebastian Scherer** and **Prof. Dr. Jean Oh** on AI for autonomous aircraft integration into manned traffic, focusing on vision-based detection and trajectory prediction.

ITACOPTER 1 — Hexacopter for Civil Construction Monitoring — Jul 2015 – Dec 2015
Aeronautics Institute of Technology (ITA) — *São José dos Campos, Brazil*
Worked with **Prof. Dr. David Antonio dos Santos** on the design and flight testing of an unmanned hexacopter for infrastructure monitoring.

Stochastic Transport Processes at Interfaces May 2015 – Aug 2015
Stony Brook University *Stony Brook, NY*
Worked with **Prof. Dr. Carlos Colosqui** on the numerical modeling of Brownian Motion and colloidal particle dynamics at micro- and nanoscale interfaces.

Automatic Data Acquisition in Plasma Tunnel Tests Aug 2012 – Jul 2013
Aeronautics Institute of Technology (ITA) *São José dos Campos, Brazil*
Worked with **Prof. Dr. Gilberto Petraconi** on development of automatic data acquisition systems for plasma environments and evaluation of thermo-structural composite materials for aerospace protective coatings.

Thermionic Cathode Discharge in Low Pressure Aug 2011 – Jul 2012
Aeronautics Institute of Technology (ITA) *São José dos Campos, Brazil*
Worked with **Prof. Dr. Gilberto Petraconi** on the experimental analysis of thermionic cathode discharge and plasma physics in low-pressure environments for ion source efficiency improvement.

PROFESSIONAL EXPERIENCE

Institute for Advanced Studies (IEAv) Jan 2016 – Present
Research Engineer *São José dos Campos, Brazil*
Research on military **Modeling & Simulation**, **Machine Learning**, and data science for decision support systems, building simulation environments and analytical tools for complex aerospace scenarios.

Brazilian Air Force (FAB) Feb 2013 – Present
Captain Engineer Officer *São José dos Campos, Brazil*
Captain (Dec 2022 – Present): AI researcher, coordinator of defense offset programs, and firearms instructor.
First Lieutenant (Dec 2015 – Dec 2022): data science research on defense scenario analysis and operational modeling.
Officer Candidate (Feb 2013 – Dec 2015): ITA research on aerial robotics, plasma physics, and advanced materials.

COC Educational System Feb 2017 – Jul 2021
Mathematics Teacher and Scientific Olympiads Coordinator *São José dos Campos, Brazil*
Orchestrated preparatory classes for Scientific Olympiads, guiding students to secure **158 medals** in 46 national and international competitions.

Emerge Brazil Oct 2016 – Dec 2017
Co-founder and Head of Communications *São Paulo, Brazil*
Bridged scientific research with industry, mapping over **300 technologies** and facilitating **60** technology-industry partnerships in the Brazilian market.

SKILLS

Natural Languages	Native Portuguese ◇ Advanced English
Programming	Python ◇ C++ ◇ R ◇ SQL ◇ MATLAB ◇ LaTeX
DL Frameworks	TensorFlow ◇ Keras ◇ PyTorch
Toolkits	NumPy ◇ Pandas ◇ Matplotlib ◇ Scikit-Learn ◇ SciPy ◇ Git ◇ Docker ◇ Linux

PUBLICATIONS

Journal Articles

[J1] Artur Ligieri Nunes, Guilherme Moreira, **Joao P. A. Dantas**, and Luiz Alberto Vieira Dias. Stopping Criteria in Simulation: A Survey of Methods, Applications, Challenges, and Future Directions. *IEEE Access*, 14:163138–163158, 2026.

[J2] **Joao P. A. Dantas**, Andre N. Costa, Diego Geraldo, Marcos R. O. A. Maximo, and Takashi Yoneyama. PoKER: A Probability of Kill Estimation Rate Model for Air-to-Air Missiles Using Machine Learning on Stochastic Targets. *The Journal of Defense Modeling and Simulation: Applications, Methodology, Technology*, 23(4):579–599, 2026.

[J3] Samara R. Silva, Vitor C. F. Gomes, Alessandro O. Arantes, Andre F. M. Caetano, Victor L. D. B. Costa, Adrisson R. Samersla, Felipe L. L. Medeiros, Yuri D. Ferreira, Marcia R. C. Aquino, and **Joao P. A. Dantas**. AsaFG: A Human-in-the-Loop Integration Module for Air Combat Simulations. *IEEE Access*, 13:155821–155834, 2025.

[J4] **Joao P. A. Dantas**, Felipe L. L. Medeiros, Adrisson R. Samersla, Pedro L. R. Botelho, Vitor C. F. Gomes, Samara R. Silva, Yuri D. Ferreira, Alessandro O. Arantes, Marcia R. C. Aquino, and Marcos R. O. A. Maximo. Deep Reinforcement Learning Agents With Collective Situational Awareness for Beyond Visual Range Air Combat. *IEEE Access*, 13:143052–143069, 2025.

[J5] **Joao P. A. Dantas**, Marcos R. O. A. Maximo, and Takashi Yoneyama. Autonomous Aircraft Tactical Pop-Up Attack Using Imitation and Generative Learning. *IEEE Access*, 13:81204–81217, 2025.

[J6] Andre N. Costa, **Joao P. A. Dantas**, Edvards Scukins, Felipe L. L. Medeiros, and Petter Ögren. Simulation and Machine Learning in Beyond Visual Range Air Combat: A Survey. *IEEE Access*, 13:76755–76774, 2025.

[J7] **Joao P. A. Dantas**, Diego Geraldo, Andre N. Costa, Marcos R. O. A. Maximo, and Takashi Yoneyama. ASA-SimaaS: Advancing Digital Transformation through Simulation Services in the Brazilian Air Force. *Spectrum: The Journal of Operational Applications in Defense Areas*, 26(1):1–7, 2025.

[J8] Mayara Condé Rocha Murça, Marcos Ricardo Omena de Albuquerque Maximo, **João Paulo de Andrade Dantas**, João Basílio Tarelho Szenczuk, Carolina Rutili de Lima, Lucas Orbolato Carvalho, and Gabriel Adriano de Melo. Open Machine Learning Models for Actual Takeoff Weight Prediction. *Journal of Open Aviation Science*, 3(2), 2025.

[J9] Jay Patrikar, **Joao Dantas**, Brady Moon, Milad Hamidi, Sourish Ghosh, Nikhil Keetha, Ian Higgins, Atharva Chandak, Takashi Yoneyama, and Sebastian Scherer. Image, speech, and ADS-B trajectory datasets for terminal airspace operations. *Scientific Data*, 12(1):468, 2025.

[J10] Ingrid Navarro, Jay Patrikar, **Joao P. A. Dantas**, Rohan Baijal, Ian Higgins, Sebastian Scherer, and Jean Oh. SoRTS: Learned Tree Search for Long Horizon Social Robot Navigation. *IEEE Robotics and Automation Letters*, 9(4):3759–3766, 2024.

[J11] Mario Viscardi, **Joao Dantas**, Diego Geraldo, and Angelo Pássaro. Advanced Defensive Tactics: Integrating Simulation and Machine Learning in Aerial Warfare. *Spectrum: The Journal of Operational Applications in Defense Areas*, 25(1):12–17, 2024.

[J12] **Joao P. A. Dantas**, Marcos R. O. A. Maximo, Andre N. Costa, Diego Geraldo, and Takashi Yoneyama. Machine Learning to Improve Situational Awareness in Beyond Visual Range Air Combat. *IEEE Latin America Transactions*, 20(8):2039–2045, 2022.

[J13] Andre N. Costa, Felipe L. L. Medeiros, **Joao P. A. Dantas**, Diego Geraldo, and Nei Y. Soma. Formation Control Method Based on Artificial Potential Fields for Aircraft Flight Simulation. *SIMULATION*, 98(7):575–595, 2022.

[J14] **Joao P. A. Dantas** and Caio Augusto de Melo Silvestre. Modelo de Simulação Aplicado às Missões de Transporte na Região Amazônica. *Spectrum: The Journal of Operational Applications in Defense Areas*, 21(1):10–15, 2020.

[J15] **Joao P. A. Dantas**, Jelton Alexandre da Cunha, Jamesson Lira Silva, Alessandro Oliveira Arantes, and Vitor Conrado Faria Gomes. Análise Exploratória de Dados de Acidentes Aeronáuticos no Brasil. *Revista Conexão SIPAER*, 9(2):106–127, 2018.

[J16] **Joao P. A. Dantas**. Teoria dos Jogos Aplicada ao Combate BVR. *Spectrum: The Journal of Operational Applications in Defense Areas*, 20:58–63, September 2017.

Conference Papers

[C1] Artur L. Nunes, **Joao P. A. Dantas**, and Luiz A. V. Dias. HaltSim: Uma Ferramenta de Avaliação para Critérios de Parada em Simulações Estocásticas. In *Simpósio de Aplicações Operacionais em Áreas de Defesa (SIGE)*, 2026.

[C2] **Joao P. A. Dantas**, Jelton A. Cunha, and Gabriel Dietzsch. Offline Multimodal Large Language Models for Decision Support in Air Operations. In *Congress of the International Council of the Aeronautical Sciences (ICAS)*, 2026.

[C3] **Joao Dantas**, Paulo Silva Filho, Jelton Cunha, and Gabriel Dietzsch. HARBOR: Heading Analysis and Reconstruction from Behavioral Observation and Radar. In *AIAA AVIATION 2026 Forum*, San Diego, CA, USA, 2026.

[C4] **Joao P. A. Dantas**. Offline Language-Based Support System for Tactical Decision-Making in Air Combat Operations. In *Proceedings of the LatinX in AI (LXAI) Workshop at the Conference on Neural Information Processing Systems (NeurIPS)*, 2025.

[C5] Francisco H. F. Araujo, Murillo S. Szvaticsek, Marco A. P. Silva, **Joao P. A. Dantas**, Christopher S. Cerqueira, and Cesar A. C. Marcondes. Análise das Formaturas Operacionais no Combate Aéreo Além do Alcance Visual para Infiltração em Território Inimigo. In *Simpósio de Aplicações Operacionais em Áreas de Defesa (SIGE)*, 2025.

[C6] Lucas S. Lima, **Joao P. A. Dantas**, and Paulo M. Tasinaffo. Algoritmo Distribuído para Enxame de Drones Defensivos de Pequeno Porte Baseado em Planejamento. In *Simpósio de Aplicações Operacionais em Áreas de Defesa (SIGE)*, 2025.

[C7] Lucas Silva Lima, Rafael Duarte Rocha, Rafael Hoffmann Giannico, Denys Derlian Carvalho Brito, and **Joao P. A. Dantas**. DroneSwarm2D: Um Simulador de Enxame de Drones Autônomos para o Estudo de Táticas Defensivas Distribuídas. In *XX Congresso Acadêmico sobre Defesa Nacional (CADN)*. Escola Superior de Defesa (ESD), 2025.

[C8] **Joao P. A. Dantas**, Andre N. Costa, Vitor C. F. Gomes, Andre R. Kuroswiski, Felipe L. L. Medeiros, Diego Geraldo, Adrisson R. Samersla, Samara R. Silva, Andre F. M. Caetano, Yuri D. Ferreira, Alessandro O. Arantes, Davison S. Santos, and Marcia R. C. Aquino. ASA: A Simulation Environment for Evaluating Military Operational Scenarios. In *Computational Science and Computational Intelligence: 11th International Conference, CSCI 2024, Las Vegas, NV, USA, Proceedings, Part XII*, Communications in Computer and Information Science, pages 203–215, Cham, Switzerland, 2025. Springer Nature Switzerland.

[C9] **Joao P. A. Dantas**, Samara R. Silva, Vitor C. F. Gomes, Andre N. Costa, Adrisson R. Samersla, Diego Geraldo, Marcos R. O. A. Maximo, and Takashi Yoneyama. AsaPy: A Python Library for Aerospace Simulation Analysis. In *Proceedings of the 2024 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (SIGSIM-PADS'24)*, Atlanta, GA, USA, 2024. Association for Computing Machinery.

[C10] **Joao P. A. Dantas**. Autonomous Pop-Up Attack Maneuver in Air Combat Using Imitation Learning. In *Proceedings of the Winter Simulation Conference, PhD Colloquium*, Orlando, FL, USA, 2024. IEEE.

[C11] **Joao P. A. Dantas**, Marcos R. O. A. Maximo, and Takashi Yoneyama. Loyal Wingman Assessment: Social Navigation for Human-Autonomous Collaboration in Simulated Air Combat. In *Proceedings of the 2024 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (SIGSIM-PADS'24)*, Atlanta, GA, USA, 2024. Association for Computing Machinery.

[C12] Lucas Silva Lima, Rafael Hoffmann Giannico, Denys Derlian Carvalho Brito, Antonio Gustavo Silveira Dantas, and **Joao P. A. Dantas**. Aprendizado de Máquina para a Otimização da Obtenção de Resultados em Simulações de Defesa Aeroespacial. In *Congresso Acadêmico sobre Defesa Nacional (CADN)*, Rio de Janeiro, RJ, Brazil, 2024. Escola Superior de Defesa (ESD).

[C13] **Joao P. A. Dantas**, Diego Geraldo, Felipe L. L. Medeiros, Marcos R. O. A. Maximo, and Takashi Yoneyama. Real-Time Surface-to-Air Missile Engagement Zone Prediction Using Simulation and Machine Learning. In *Interservice/Industry Training, Simulation and Education Conference (I/ITSEC)*, Orlando, FL, USA, 2023. National Training and Simulation Association (NTSA).

[C14] Gabriel Henrique Gobi, Pedro Lustosa Rege Botelho, Thiago Lobo Ferreira, Thiago Lopes Araujo, and **Joao P. A. Dantas**. Ambiente de Simulação Aeroespacial: Capacidades e Potenciais Benefícios para a Indústria de Defesa Brasileira. In *Congresso Acadêmico sobre Defesa Nacional (CADN)*, Pirassununga, SP, Brazil, 2023. Escola Superior de Defesa (ESD).

[C15] **Joao P. A. Dantas**, Marcos R. O. A. Maximo, and Takashi Yoneyama. Autonomous Agent for Beyond Visual Range Air Combat: A Deep Reinforcement Learning Approach. In *Proceedings of the 2023 ACM SIGSIM Conference on Principles of Advanced Discrete Simulation (SIGSIM-PADS'23)*, Orlando, FL, USA, 2023. Association for Computing Machinery.

[C16] **Joao P. A. Dantas**, Andre N. Costa, Felipe L. L. Medeiros, Diego Geraldo, Marcos R. O. A. Maximo, and Takashi Yoneyama. Supervised Machine Learning for Effective Missile Launch Based on Beyond Visual Range Air Combat Simulations. In *Proceedings of the Winter Simulation Conference (WSC'22)*, Singapore, 2022.

[C17] Jay Patrikar, **Joao P. A. Dantas**, Sourish Ghosh, Parv Kapoor, Ian Higgins, Jasmine J. Aloor, Ingrid Navarro, Jimin Sun, Ben Stoler, Milad Hamidi, et al. Challenges in Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace. In *Aerial Robotics Workshop, International Conference on Robotics and Automation (ICRA)*, 2022.

[C18] **Joao P. A. Dantas**, Andre N. Costa, Diego Geraldo, Marcos R. O. A. Maximo, and Takashi Yoneyama. Weapon Engagement Zone Maximum Launch Range Estimation Using a Deep Neural Network. In *Intelligent Systems (BRACIS)*, Lecture Notes in Computer Science, pages 193–207, Cham, 2021. Springer.

[C19] **Joao P. A. Dantas**, Andre N. Costa, Diego Geraldo, Marcos R. O. A. Maximo, and Takashi Yoneyama. Engagement Decision Support for Beyond Visual Range Air Combat. In *Proceedings of the 2021 Latin American Robotics Symposium, 2021 Brazilian Symposium on Robotics, and 2021 Workshop on Robotics in Education (LARS/SBR/WRE)*, pages 96–101, Natal, RN, Brazil, 2021. IEEE.

[C20] **Joao P. A. Dantas**, Andre N. Costa, Marcos R. O. A. Maximo, and Takashi Yoneyama. Enhanced Self-Organizing Map Solution for the Traveling Salesman Problem. In *Anais do XVIII Encontro Nacional de Inteligência Artificial e Computacional (ENIAC)*, pages 799–802. SBC, 2021.

[C21] **Joao P. A. Dantas** and Gilberto Petraconi Filho. Aquisição Automática de Dados em Ensaios de Materiais de Barreira Térmica Realizados em Túnel de Plasma Supersônico. In *Anais do XIX Encontro de Iniciação Científica e Pós-Graduação do ITA (XIX ENCITA)*, 2013.

[C22] **Joao P. A. Dantas** and Gilberto Petraconi Filho. Estudo de uma Descarga de Catodo Termiônico em Baixa Pressão. In *Anais do XVIII Encontro de Iniciação Científica e Pós-Graduação do ITA (XVIII ENCITA)*, 2012.

Theses

[T1] **Joao P. A. Dantas**. *Simulation and Machine Learning for Decision Support and Autonomy in Air Combat Operations*. PhD thesis, Aeronautics Institute of Technology, 2025.

[T2] **Joao P. A. Dantas**. Apoio à Decisão para o Combate Aéreo Além do Alcance Visual: Uma Abordagem por Redes Neurais Artificiais. Master's thesis, Instituto Tecnológico de Aeronáutica, São José dos Campos, SP, Brazil, 2018.

[T3] **Joao P. A. Dantas**. Hexacóptero para Monitoramento de Construção Civil: Montagem, Testes e Operação. Senior thesis, Instituto Tecnológico de Aeronáutica, São José dos Campos, SP, Brazil, 2015.

Preprints

[P1] Andre N. Costa and **Joao P. A. Dantas**. Analysis of Sequential Parameter Optimization for Computer Simulation Optimization. *SSRN eLibrary*, 2023.

SOFTWARE

Dec 2025: DroneSwarm2D — Autonomous drone swarm simulator for distributed defensive tactics. (INPI, 2025).
Dec 2025: AsaGym — Reinforcement Learning library for aerospace scenarios evaluation. (INPI BR 51 2026 003324-1).

Dec 2025: AsaFG — Human-in-the-loop integration module for air combat simulations. (INPI BR 51 2026 003322-5).
Oct 2024: ASA (Ambiente de Simulação Aeroespacial) — Simulation framework for modeling military scenarios. (INPI BR 51 2024 002174-4).
Jun 2023: AI Pilot — Close-proximity safe and seamless operation of manned and unmanned aircraft in shared airspace. (CMU Tech Transfer 2023-102).
2025: TartanAviation — Large-scale terminal airspace operations datasets.
2024: AsaPy — A Python library for aerospace simulation analysis and visualization.

SUPERVISION

1. **Artur Ligieri Nunes**. Stop Criteria in Defense Simulations: Comparative Analysis and Statistical Approach Proposal. 2025. M.Sc. Thesis (Electronic and Computer Engineering) — Aeronautics Institute of Technology. (*Ongoing*).
2. **Lucas Silva Lima**. Proposta de Algoritmo Distribuído para Enxame de Drones Defensivos de Pequeno Porte Baseado em Inteligência Artificial. 2025. M.Sc. Thesis (Electronic and Computer Engineering) — Aeronautics Institute of Technology.
3. **Lucas Silva Lima**. Proposta de Algoritmo Distribuído para Enxame de Drones Defensivos de Pequeno Porte Baseado em Planejamento. 2025. B.Sc. Project (Computer Engineering) — Aeronautics Institute of Technology.
4. **Victor Luiz Dorea Beirão Costa**. Apoio à Decisão no Planejamento Baseado em Capacidades: Uma Análise de Cenários de Defesa no Ambiente de Simulação Aeroespacial. 2024. B.Sc. Project (Mechanical-Aeronautical Engineering) — Aeronautics Institute of Technology.
5. **Gabriel Henrique Gobi** and **Pedro Lustosa Rege Botelho**. AsaGym: A Python Library for Reinforcement Learning in Aerospace Operational Scenarios. 2023. B.Sc. Project (Computer and Electronic Engineering) — Aeronautics Institute of Technology.

HONORS & RECOGNITIONS

Dec 2025: 3rd place in the Simulation Challenge at the IEEE Winter Simulation Conference 2025 ($1,000).
Nov 2025: 1st place in the Brazilian Ministry of Defense Doctoral Award 2025 for the best Ph.D. thesis in defense science and technology (R$4,000).
Dec 2024: WSC Diversity Award at the IEEE Winter Simulation Conference 2024 in Orlando, USA.
Dec 2024: Runner-up Best Poster Award at the LACORO Summer School on Robotics 2024 in Chile.
Nov 2024: 2nd place in the Performance Review Commission (PRC) Data Challenge 2024 with an open Machine Learning model to infer flight Actual TakeOff Weight (ATOW).
Jun 2024: ML and Systems Rising Star 2024, awarded by MLCommons.
Oct 2023: 3rd place in the Data Science Challenge 2023 for predicting the Estimated Landing Time (ELDT) of flights.
Jun 2023: Best PhD Colloquium Award at the Conference on Principles of Advanced Discrete Simulation (PADS) 2023.
Oct 2021: 3rd place in the 5th Brazilian Competition on Knowledge Discovery in Databases (KDD-BR), predicting solutions of the Travelling Salesman Problem (R$200).
Oct 2021: Military Bronze Medal, awarded by the Brazilian Air Force.
Dec 2019: Ambassador Award of Excellence 2019, International Youth Math Challenge (IYMC).
Oct 2019: 6th place (39 competitors) in the 3rd Brazilian Competition on Knowledge Discovery in Databases (KDD-BR), predicting flight delays for a Brazilian airline.
May 2019: 4th place in the Data Science Challenge at Engineering Education for the Future (EEF) 2019, predicting real estate prices.
Dec 2015: Honorable Mention in the Department of Humanities, Aeronautics Institute of Technology.
Mar 2014: Military Merit Award, Brazilian Air Force (top 25% behavioral grades).
Nov 2011: 1st place overall in the Brazilian Air Force Reserve Officer Training Corps (CPORAER-SJ) (120 candidates).
Dec 2010: ITA Entrance Exam Approval at the Aeronautics Institute of Technology (120 admitted out of 7,627 applicants), one of Brazil's most competitive entrance exams.
Jan 2009: Academic Merit Award, Brazilian Air Force (top 25% military grades).
Oct 2007: Bronze Medal in the 10th Brazilian Olympiad of Astronomy and Astronautics (OBA) (187,726 competitors).
Dec 2006: Honorable Mention in the XIV Ceara Science Olympiad — Biology (OCC), Federal University of Ceara (UFC).
Dec 2006: Gold Medal in the Internal Science Olympiads 2006 (Math, Physics, Biology), Farias Brito High School.
Nov 2003: Honor to Merit (three 2nd places, one 5th place) in the Internal Mathematics Olympiad, Christus Middle School.
Oct 2003: 2nd place in the 3rd Christus Exhibition of Science and Technology (EXCETEC), category "Exact, Social and Commercial Sciences".

FELLOWSHIPS & GRANTS

May 2026: DAAD AInet Fellow (AI in Neuroscience), German Academic Exchange Service (DAAD), Postdoctoral Networking Tour in Artificial Intelligence (Postdoc-NeT-AI).
Dec 2025: Travel Grant ($1,000) to attend the IEEE Winter Simulation Conference 2025 in Seattle, USA.
Dec 2025: Travel Grant ($1,500), LatinX in AI Coalition, for the LatinX in AI Research Workshop at NeurIPS 2025, San Diego, USA.

Jun 2025: DAC Young Fellow for the Design Automation Conference (DAC) 2025.
Jun 2025: Travel Grant (€2,900), CISB Call 15/2025, for the 2025 Swedish Aerospace Congress.
Dec 2024: Travel Grant ($1,000) to attend the IEEE Winter Simulation Conference 2024 in Orlando, USA.
Dec 2024: Travel Grant ($500) to attend the LACORO Summer School on Robotics 2024 in Chile.
Jun 2024: Travel Grant ($1,000) to attend the 2024 Conference on Principles of Advanced Discrete Simulation (PADS).
May 2024: Travel Grant ($3,000) to attend the UNODA-SIPRI Workshop on Responsible AI for Peace and Security in Brussels.
May 2024: Travel Grant (€3,150), CISB Call 13/2024, for the WARA-PS Demonstration Week 2024.
Jun 2023: Travel Grant (€3,200), CISB Call 10/2022, for an AI researchers delegation in Sweden.
Jun 2023: Travel Grant ($1,000) to attend the Conference on Principles of Advanced Discrete Simulation (PADS) 2023 in Orlando, USA.
Apr 2023: Travel Grant ($2,500) to attend Brazil at Silicon Valley 2023 in Santa Clara, USA.
Jun 2022: Merit Scholarship, Fundação Estudar, awarded for outstanding trajectory and academic potential (30 of 33,876 applicants).
Oct 2016: Prolider Program Fellow 2016, selected by Instituto Four (29 of 2,013 applicants).
Aug 2014 – May 2015: Full Scholarship (≈$75,000) from Brazil's Ministry of Education for a one-year study program at Stony Brook University, College of Engineering.
Aug 2012 – Jul 2013: Undergraduate Research Fellowship (PIBIC), CNPq, at the Aeronautics Institute of Technology, for the project "Automatic Data Acquisition in Plasma Tunnel Tests".
Aug 2011 – Jul 2012: Undergraduate Research Fellowship (PIBIC), CNPq, at the Aeronautics Institute of Technology, for the project "Thermionic Cathode Discharge in Low Pressure".
Jan 2004 – Dec 2006: Full Scholarship (≈$30,000) for three years at Farias Brito High School.

VOLUNTEERING

Fundação Estudar Jan 2025 – Dec 2025
Ambassador *São Paulo, Brazil*
Engaging with a network of high-achieving young leaders, supporting initiatives to promote education and career development for Brazilian students.

International Astronomy and Astrophysics Competition Apr 2020 – Jul 2021
Ambassador *São José dos Campos, Brazil*
Inspiring youths for astronomy and astrophysics and encouraging participation in the competition.

International Youth Math Challenge Jul 2019 – Jul 2021
Ambassador *São José dos Campos, Brazil*
Informing schools and encouraging students and youths to participate in the competition.

Alpha Lumen Institute Jun 2012 – May 2013
Mathematics Teacher *São José dos Campos, Brazil*
Teacher of mathematics in the preparatory class for the ITA entrance exam.

OPERATIONAL COURSES

- **Emergency Parachute Jump Training (ISE)** – Brazilian Air Force (2008).
- **Firearms Instructor Course** – Brazilian Air Force (2023).
- **Open Water Diver Course (Certification No. 26030I6411)** – Professional Association of Diving Instructors (PADI) (2026).
- **Military Remotely Piloted Aircraft Operator Course** – Brazilian Air Force (2026).

ADDITIONAL ACTIVITIES

- **Sports:** Basketball player at Air Force Preparatory School and Air Force Academy (2007–2009). Amateur bodybuilder (2014–2015) – The National Physique Committee. Active Brazilian Jiu-Jitsu practitioner (blue belt, IBJJF).
- **Outdoors and Interests:** Orienteering, trekking, reading (science fiction), chess, and traveling.

November 7, 2026